Reg. No. | | | | | | | | | | | [17CS0SP02]

# FOURTH SEMESTER B.TECH EXAMINATION, JUNE - 2019
# RELATIONAL DATABASE MANAGEMENT SYSTEM

**Time: 3 Hours** **Maximum Marks:70**

**Instructions:**

1. Answer any **FIVE** full questions. Choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed.

## PART-A

**1. a.** Consider two friends A and B have enormous of data in mobile. They want the organized data in the mobile to be stored in database and approached the database company C. Elaborate on the salient features must be provided by the company C to their clients. **08**

**b.** Describe the three schema architecture. **06**

**2. a.** With a neat diagram, explain the main components of the database management system and their interactions. **08**

**b.** Consider the schema **06**

**STUDENT**(Name, Student_number, Class Major)
**COURSE(**Course_name, Course_number, Credit_hours, Department)
**PREREQUISITE(**Course_number, Prerequisite_number)
**SECTION(**Section_identifier, Course_number, Semester, Year, Instructor)
**GRADE_REPORT(**Student_number, Section_identifier, Grade).
Create the network model.

**3. a.** Consider an employee X have been appointed in the database company Y. The employee allotted to the client Z to support for the design of application. With the necessary diagram, explain the fundamental steps that an employee X has to adapt during the database design process. **08**

**b.** Consider the scenario with the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course: **06**

STUDENT(<u>Ssn</u>, Name, Major, Bdate, age, {phonenumber}, address(Apt#, street#, zip))
COURSE(<u>Course#</u>, Cname, Dept)
ENROLL(<u>Ssn</u>, <u>Course#</u>, <u>Quarter</u>, Grade)
BOOK_ADOPTION(<u>Course#</u>, <u>Quarter</u>, Book_isbn)
TEXT(<u>Book_isbn</u>, Book_title, Publisher, Author).
Identify the different types of attributes.

**4. a.** Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies $F = \{ \{A, B\}\rightarrow\{C\}, \{A\}\rightarrow\{D, E\}, \{B\}\rightarrow\{F\}, \{F\}\rightarrow\{G,H\}, \{D\}\rightarrow\{I, J\} \}$. What is the key for $R$? Decompose $R$ into 2NF and then into 3NF relations. **08**

**b** Demonstrate how INSERT and DELETE operations can violate the referential integrity constraint. Recommend the remedial measures to overcome the problem. **06**

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## PART-B

5. a. Consider the following relations COMPANY relational database schema in SQL **08**

**EMPLOYEE** ( Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno)

**DEPARTMENT** (Dname, Dnumber, Mgr_ssn, Mgr_start_date)

**DEPT_LOCATIONS** (Dnumber, Dlocation)

**DEPENDENT** (Essn, Dependent_name, Sex, Bdate, Relationship)

Build the SQL queries for the following

i. Retrieve the birthdate and address of the employee whose name is 'John B. Smith'.

ii. Retrieve the name and address of all employees who work for the 'Research' department.

iii. For each employee, retrieve the employee's name, and the name of his or her immediate supervisor.

iv. Retrieve the name of each employee who has a dependent with the same first name as the employee.

b. The DBA working in a company X has created the two users user1 and user2 with create session and create table permissions. The user1 created the database application for storing the student marks information and user2 created the application for storing the faculty information. Since both the users left the company the DBA revoked the permissions from the both the users. Create the SQL TCL commands to demonstrate the above scenario. **06**

6. a. What is SQLJ iterator? Explain the different types of iterators. **05**

b. Distinguish between JDBC and SQLJ. Why do they both exist? **05**

c. Outline the Typical Sequence of Interaction in Database Programming. **04**

7. a. Demonstrate the different types of loop control structures supported in PL/SQL. **08**

b. Build the PL/SQL program to find the factorial of the given number. **06**

8. a. Create a **row-level** trigger for the customers table that would fire for INSERT or UPDATE or DELETE operations performed on the CUSTOMERS table. This trigger will display the salary difference between the old values and new values. CUSTOMER schema given below. **05**

Customers(ID | NAME | AGE | ADDRESS | SALARY).

b. Create a PL/SQL function that returns the tax of the person based on the annual salary input. The tax computation is given below. **05**

If salary between 0 & 2000, 10 percent is deducted as tax.

If salary between 2001 & 4000, 15 percent is deducted as tax. If salary above 4000, 25 percent is deducted as tax.

c. Recommend the solution using PL/SQL package to find the salary of the given customer. CUSTOMER schema given below. **04**

Customers(ID | NAME | AGE | ADDRESS | SALARY).